

Machine Translation:-

* It is a process of converting the text from one language to other automatically without or minimal human intervention.

Eg: India is a very great country - English
 भारत है एक महान देश - Marathi

Why to study machine Translation?

- ① one of the most challenging problem in NLP
- ② pushes the boundaries of NLP
- ③ Involves analysis as well as synthesis.
- ④ Involves all layers of NLP: morphology, syntax, semantics, pragmatics, discourse.
- ⑤ Involves analysis as well as synthesis.
- ⑥ Theory & techniques in MT are applicable to a wide range of other problems like transliteration, speech recognition & synthesis.

* Challenges in Machine Translation:-

1) Topology: It is the scientific study of cross-linguistic similarities & contrasts.

a) morphological:-

- Categorized morphologically along two dimensions of variation.
- The first is the number of morphemes per word.
- 2nd factor is how easily morphemes can be divided

from agglutinative languages like Turkish having relatively clear morpheme borders to fusion languages like Russian where a single affix can combine several morphemes.

b) Syntactic: -

Languages that tend to place verb between the subject & the object include German, French, English & Mandarin for instance.

Eg: Ram (S) like(s) ice-cream (O).

c) Head marking & dependent marking: -

Languages that mark the head often mark the relationship between the head & its dependents.

Language that mark dependencies frequently mark the relation on the non-head.

d) Verb framed & Satellite framed: -

Languages can be distinguished by whether the verb or one of the satellites - particles, prepositional phrases or adverbial phrases - marks the direction & mode of motion.

Verb framed languages, such as Spanish, which has the verbs *acercarse* (approach, reach, enter & exit), signal the direction of motion on the verb leaving the satellites to mark the mode of motion.

② Referential Density: -

When referring to a specific referent in the discourse, many languages demand that we employ an explicit pronoun.

2) Other structural divergences: -

- Topological distinctions are the basic for many structural variations between languages.
- Others are only quirky variations unique to certain languages or language combinations. Adjectives often come after nouns in French & Spanish phrase places adjectives before nouns.

Eg: English : Green Ball
 Marathi : ~~हिरवा~~ (Green)

3) Lexical Divergence: -

- It provides significant challenges for translators, when words have multiple senses (meaning) the translation becomes difficult.

Even in cases of polysemy, however we often have to disambiguate if the target language doesn't have the exact same kind of polysemy.

Eg: The English polysemous word 'know' it can be used to describe having knowledge of a truth or idea or having familiarity with

Someone or somewhere.

Fact - I 'know' Ram

Hindi uses two words for the same English word 'know'

Fact - I 'know' snow is white ^{हम} ^{को} ^{हम}
 कि यह सच है !

Familiarity - I 'know' Ram - मैं राम को जानता हूँ।

* Machine translation techniques categorized into 4 types

- Rule based Machine Translation.
- Statistical MT
- Hybrid MT
- Neural MT

* Rule-Based Machine Translation:-

The Earliest Commercial MT systems were rule-based machine translation (RBMT) systems which are based on linguistic principles that permit words to be placed in many contexts & to have various meanings.

- Large collections of linguistic rules can be applied to using RBMT technology in 3 stages: analysis, transfer & generation.
- Rules are created by programmers & human language experts who have put a lot of work into understanding the manual construction of

Some of the lexicons used by RBMT allows users to update & improve the translations.

* Advantages:—

- No bilingual texts are required
- Domain independent
- No quality ceiling
- Total Control
- Reusability

* Disadvantages:—

- Insufficient amount of really good dictionaries, Building new dictionaries is expensive.
- o Some linguistic information still needs to be set manually.
- o It is hard to deal with rule interactions in big systems ambiguity & idiomatic expressions.
- o Failure to adapt to new domains.

* Statistical Machine Translation (SMT):—

- o A machine translation paradigm known as Statistical machine translation (SMT) bases translation on statistical models, the parameters of which are generated from analysis of sizable bilingual text corpora. The collection of a sizeable & organized

group of writings written in two different languages is referred to as a bilingual text corpus.

- Information theory, which examines the qualification, storage & transmission of information is the main foundation of SMT.

- To create the statistical models, supervised & unsupervised machine learning techniques are employed.

* Cross lingual Translation: —

* Sentiment Analysis: —

- It is often known as opinion mining, is a technique used in natural language processing to determine the emotional undertone of a document.
- This is a common method used by organisations to identify & group ideas regarding a certain good service or concept.
- It can be extract the subject, opinion header & polarity from text in addition to identifying sentiment.

7

* Challenges in Sentiment analysis:

① Subjectivity & Tone

Just as crucial as determining the tone of a piece of identifying its subjective & objective components.

Eg: Considering two paragraphs:

The packaging seems good

The container is red

Most would agree that the sentiment is favourable for the 1st sentence than the second.

② Context & polarity:

- Every word is spoken at some point in time, some-where, by some^{one} & to some one
- Every word is said in its proper context:
- Without context, analysing sentiment becomes very challenging.

③ Irony & Sarcasm:

Without a detailed group of the situation in which a feeling was expressed, robots may struggle to recognise irony & sarcasm because people often communicate their bad feelings using pleasant phrases.

④ Comparison: — How to handle comparisons:

Eg: This product is second to none.

This is better than older tools.

This is better nothing.

The identification of the sentiment for the 1st sentence is quite straight forward, it is positive. However, 2nd & 3rd sentence is bit more challenging.

⑤ Emoji's:-

When conducting sentiment analysis on tweets, particular attention must be paid to character-level as well as word-level emoji's. Extensive preprocessing may be required.

⑥ Defining Neutral:-

It is based on perception, i.e. the perception of the engineer defining the which words or sentiment words might be different than those of the customers giving the review.

* Types of Sentiment analysis:-

1) Fine-grained:-

Sentiment are divided into additional categories, typically highly positive to very negative, to provide a more exact level of polarity.

This can be compared to ratings on a 5 Star Scale in terms of opinion.

2) Emotion detection:— Instead of identifying positivity & negativity, emotion detection recognises particular emotions.

Eg: Shock, rage, grief, frustration & happiness.

3) Intent based:—

Analysis is performed with an eye toward intent sees more than just the author's opinion.

4) Aspect based:— It collects the precise component that is being monitored either favourably or unfavourably.

* Steps in sentiment Analysis:—



Fig. Sentiment Analysis Steps:

1) Data Collection:—

Initial stage in sentiment analysis. The gathering of information from websites such as user forums, twitter, facebook, blogs & commercial portals like Amazon.com & Alibaba.com others.

2) Text preparation:— prior to analysis, text preparation reviews the data.

Some reviews & conversations on social media platforms use unpleasant & inappropriate language, thus these are investigated & prepared for a more accurate analysis.

* The goal is to eliminate spam & unsuitable reviews before sending them for automatic analysis.

• POS techniques which are based used to prepare texts for analysis.

3) Sentiment Detection:— Also known as opinion mining (OM) newline & sentiment analysis, is the act of identifying the sentiment newline represented in a person's viewpoint using machine learning or NLP.

4) Sentiment classification:— It involves extracting the text's meaning & categorizing it according to the polarity it includes, such as whether they are ve or ve, good or bad or terrible, or if they are liked or disliked.

It can be categorized into 3 types

- lexicon based approach
- machine learning approach
- Hybrid approach.

* Advantages:

- 1) Up Selling opportunities
It is easier to identify chatlers who could be open to making large purchases & helps you avoid upsetting, unhappy customers with any unwanted sales efforts.
- 2) Agent monitoring
- 3) Training chatbots
- 4) Identifying key emotional triggers.
- 5) Handling multiple customers.
- 6) Adaptive customer service.
- 7) Live Insights
- 8) Quick Escalations
- 9) Reduce customer churn

* Question Answering:

Building systems that automatically respond to - questions represented by humans in Natural languages is the focus of the computer-science topic of question answering (QA), which falls under the umbrella of information retrieval & NLP.

QAs consists of 3 parts

- a) Question processing
- b) Information retrieval
- c) Answer processing.

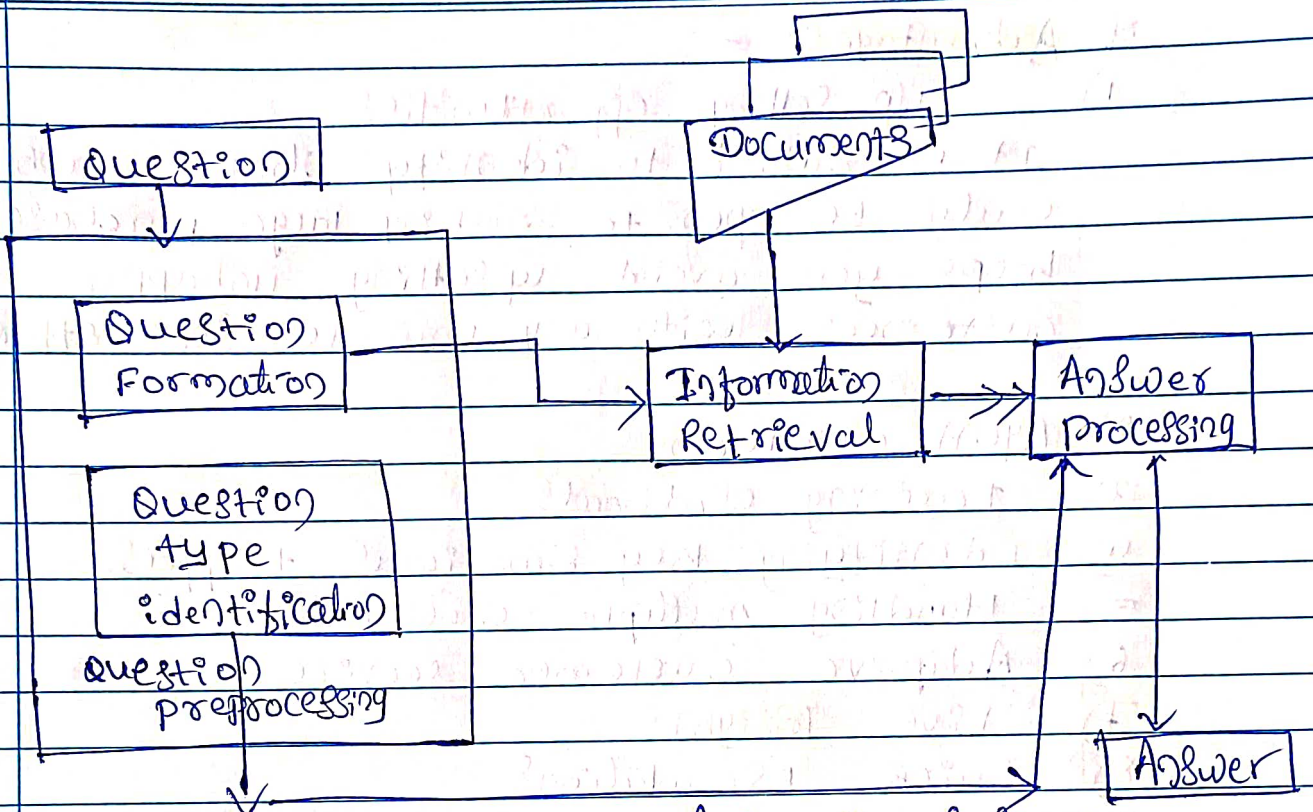


Fig. Question Answering System.

1) Question processing: - It is the 1st step in the QAS is processing the questions posted by the users.

- Users ask queries using everyday languages
- Once the inquiries have been processed, the user's true intentions can be discerned.
- The option of question type identification & inquiry formulation are carried out separately.

* Question formulation: - This method involves choosing the appropriate terms from the query.

- In accordance with IR s/m that has been chosen the keywords must be produced.

(13)

* Question Type Identification:-

While processing answers, the type of question can be used to determine which response is the most relevant.

- Machine learning & deep learning classification techniques can be applied to the type identification problem.

2] Information Retrieval:-

It's primary duty is to locate & gather all the pertinent data it can for the given topic.

- The o/p of the question formulation stage is the question processing phase serves as the i/p to IR system.

3] Answer processing:-

- It is the last stage in QAS.
- In this, using these piece of information, a potential answer is recognised & sent to the user after the question processing phase receives the type of the question & IR stage identifies any viable answer documents or answer passages.

* Challenges / Issues in the development of QAS:-

(1) Lexical gaps:-

When a QAS machine that have been taught using a specific vocabulary comes

across queries that are presented using a different vocabulary it becomes challenging for the system to accurately determine the correct answer for the given query.

② Ambiguity:

A QAS may run into ambiguity while processing questions from users as well. The QAS may run into ambiguity at the information retrieval stage as well.

- The system would not be able to extract pertinent information if the data or the knowledge itself is ambiguous, rendering the QAS ineffective.

③ Complex data:

A QAS fails if there are any inconsistencies in the knowledge it receives.

- It is envisaged that a QAS will be given highly accurate & unambiguous facts as knowledge.

u) Complex queries:

The system can be accurately evaluate whether a question is straightforward & simple while processing it. However the queries get more challenging when the user wants to know several facts.

5) partial knowledge & incorrect knowledge.

The creator & designers of the QAS would have to take into consideration is partial knowledge, incomplete knowledge & incorrect knowledge.

* Text Entailment:

- In NLP, a directional relationship between text fragment is known as textual entailment (TE). When the truth of the one text fragment follows from another text, the relationship is valid.

The Entailing & entailed texts are referred to as text (t), hypothesis (h) respectively in the TE framework.

- If a typical human reading t would conclude that h is most likely true, then "t entails h" (th) applies.

Even while "t entails h" the inverse "h entails t" is far less strong. Hence the relationship is directional.

An eg. of a positive TE.

⇒ Text: If you help the needy, God will reward you.

Hypothesis: Giving money to a poor man has good consequences.

An Eg. of Negative TE

Text: If you help the needy, God will reward you.

Hypothesis: Giving money to a poor man has no consequences.

An Eg of Non-TE

Text: If you help the needy, God will reward you.

Hypothesis: Giving money to a poor man will make you a better person.

* Discourse processing:

→ Language often consist of Collocated, connected groupings of sentences rather than single, unrelated utterances. Such a collection of sentence is referred to as a discourse.

- Language in use is what the word discourse in linguistics refers to discourse processing is the practice of analysing texts or languages that entails understanding social interactions, & text interpretation.

Dealing with morphemes, n-grams, tenses, linguistic features, page layouts & other elements can be part of discourse analysis.

- There are 2 major functions in discourse processing reference & coreference resolution.

* Dialog and Conversational Agents:-

- Dialog Systems are systems which can converse with a human any natural language. This conversation can be via speech or through text.
- Ideally a chatbot system is a simple conversation with the human. These normal & basic conversations are also called as Small Talk.

• Categories of Dialogue Systems:-

Technically conversational agents or dialogue systems can be categorised into different types as well.

- 1) Based on the way the dialogue is controlled. The way of the dialogue is controlled can be categorised into 3 types.

- a) Finite state based systems.
- b) Frame based systems.
- c) Agent based systems.

o) Finite State based System:-

There are finite states through the user communication is directed in order to have a dialogue.

Eg: In order to order a pizza, the 1st step is check for the type of pizza.

then comes the type of crust then comes the toppings, the side orders & then finally the payment for the pizza.

All steps are followed. The dialogue will fail if any of the states are skipped.

(b) Frame based system it tries to fill certain gaps in order to perform the conversation.

(c) Agent based system:-

These type of dialogue systems are more user friendly & it can be said these are the actual artificially intelligent systems capable of engaging the users in a better way compared to the state based or the frame based systems.

2) Based on the way communication is carried out between system & the user.

Can be categorised into 4 types:

Text-based, spoken dialogue based, graphical user interface & multi-modal.

In text based systems the communication between the user & the system is through typed textual format.

- Spoken dialogue based systems are much more difficult to implement & to work with.

The reason for this is first & foremost the system has to process the speech & family has to answer in the spoken format as well. Both these things are difficult & challenging.

- Dialogue systems are the systems which communicate with the users using the graphical user interfaces.
- Multi-modal system provides the user with multiple ways through which communication can be established & continued.

3) Based on the work done by the dialogue system:

Based on the work undertaken by a conversational agent can be categorised in 2 types -

- a) Task oriented dialogue
- b) chatbot systems.

- Task based systems are dialogue systems which use the user conversations as basis to perform certain tasks.

These tasks can be anything like controlling an appliance or setting up reminders or setting up calls or finding directions or search for a hotel.

- chatbot systems are simple conversational systems designed to continue the conversation with the users. They are designed in such a way

that give a sense of human touch while having a conversation with the users.

* Natural Language Generation:-

NLG includes computational linguistics, natural language processing (NLP) & natural language understanding & is connected to human-to-machine & machine-to-human interaction (NLU).

* Working of NLG:-

The NLG's six stages:-

① Content Analysis:-

Filtering data & identifying the important content.

② Data understanding:-

Interpreting data, identifying pattern & putting it into context.

③ Document structuring:-

Creating a document plan & narrative structure.

4) Sentence aggregation:-

combining relevant sentences to summarize the topic

5) grammar structuring:- Applying grammatical rules to create natural-sounding text.

⑥ Language presentation:-

Formatting & presenting final output.

Uses:-

- Producing the responses of voice assistants & chatbots like Apple's Siri & Google's Alexa.
- Automating chat, email & message responses for lead nurturing.
- Customising reactions to emails and conversations from customers.
- Creating & customising customer service employees' scripts.
- Compiling & condensing news reports.
- Reporting on how internet of things devices are doing.
- Composing product descriptions for customer communications & e-commerce websites.
- Creating material that is simple to understand for both staff & customers out of financial reports & other sorts of business data.